



Pearls AQI Predictor (Karachi)

Project Report

Muhammad Rayan

1. Executive Summary

The Pearls AQI Predictor is a fully server-less end-to-end machine learning system designed to forecast Karachi’s Air Quality Index (AQI) for the next **24, 48, and 72 hours**.

The system automates data ingestion, feature engineering, model training, experiment tracking, model registration, and deployment through a Streamlit dashboard. It operates entirely using free-tier services and GitHub Actions for orchestration.

2. Project Objectives

- Build a server-less ML pipeline for AQI forecasting
- Automate data collection and feature updates (hourly)
- Train and compare multiple models for different forecast horizons
- Track experiments and manage models using a registry
- Deploy a Streamlit dashboard for real-time predictions and insights

3. System Architecture Overview

The system is composed of the following modular components:

Layer	Component	Technology
Data Ingestion	API Integration	Open-Meteo APIs
Feature Store	Version-controlled dataset	GitHub-hosted Parquet file
Orchestration	Hourly + daily workflows	GitHub Actions
Training Pipeline	Multi-model training & evaluation	Ridge / Random Forest / XGBoost + MLflow

Layer	Component	Technology
Model Registry	Experiment tracking & model storage	MLflow on DagsHub
Serving Layer	Real-time forecast dashboard	Streamlit

4. Data Sources

All data is sourced from **Open-Meteo APIs**.

4.1 Air Quality API

Provides hourly pollutant measurements:

- PM2.5, PM10
- CO, NO₂, SO₂, O₃
- Dust, AQI

4.2 Weather API

Provides meteorological features:

- Temperature, humidity
- Wind speed & direction
- Pressure, solar radiation
- Boundary layer height
- Vapour pressure deficit

4.3 Data Processing

- Data is merged on timestamp (unix_time)
- Deduplication is applied per city + timestamp
- Stored in **data/features.parquet**

5. Pipeline Design

5.1 Backfill Pipeline

- Performs one-time historical data collection (90 days)
- Builds the initial dataset
- Output: **features.parquet**

File: backfill_pipeline.py

5.2 Hourly Feature Pipeline

- Runs every hour via GitHub Actions

- Fetches latest data from Open-Meteo
- Deduplicates and appends to dataset
- Pushes updated Parquet file to GitHub

File: feature_pipeline.py

Workflow: hourly_feature_pipeline.yml

5.3 Training Pipeline

- Loads dataset from Parquet store
- Supports two modes: full historical training (90 days) and incremental daily retraining (last 24 h focus)
- Trains multiple models: **Ridge Regression, Random Forest, XGBoost**
- Wraps models using **AQIDeltaRegressor**
- Logs metrics to MLflow and registers best model per horizon

File: training_pipeline.py

5.4 Model Registry

- Hosted on MLflow (DagsHub)
- Stores model versions, metrics, and artifacts
- Used for production model selection

5.5 Feature Store Strategy

- No external feature store used (free-tier limitation)
- GitHub-hosted Parquet file acts as feature store
- Updated automatically via CI pipeline

5.6 Dashboard (Serving Layer)

- Built with Streamlit
- Loads latest registered models from MLflow
- Outputs forecasts for: **24 h, 48 h, 72 h**
- Displays feature importance (SHAP-style bar plot)
- Includes AQI alert system (threshold > 150)

File: app.py

5.7 EDA Module

- Generates correlation heatmaps, feature distributions, and target distribution plots
- Saves outputs to **/plots**

File: eda.py

6. Model Performance

Forecast Horizon	Best Model	RMSE	MAE	R ² (OOF)	R ² (Final Fold)
24 hours	XGBoost	15.33	10.88	0.276	0.656
48 hours	XGBoost	21.40	15.34	-0.304	0.180
72 hours	XGBoost	22.23	16.90	-0.407	-0.198

Key Observations

- 24 h model performs well, capturing meaningful signal in the data
- 48 h and 72 h models underperform, indicating limited predictive power at longer horizons
- Performance degradation is expected due to: limited dataset size (90 days) and surface-level features only

7. CI/CD Automation

Two GitHub Actions workflows power the system:

Workflow	Purpose	Actions
Hourly Pipeline	Feature store freshness	Updates feature store every hour
Daily Training Pipeline	Automated model retraining	Retrains models, logs to MLflow, updates registered models

8. Challenges & Solutions

Challenge	Problem	Solution
Lack of Free Feature Store / Registry	Hopsworks and Vertex AI required payment details	MLflow on DagsHub for model registry; GitHub-hosted Parquet as feature store
Parquet Merge Conflicts	Concurrent updates caused conflicts	CI pipeline made sole owner of dataset updates
Weak Long-Horizon Performance	Poor results for 48 h and 72 h predictions	Documented as a limitation; 24 h forecast kept as primary usable output

9. Future Improvements

- Replace Parquet-in-Git with a proper cloud feature store
- Add external datasets: traffic density, satellite aerosol data
- Explore advanced models: LightGBM, LSTMs / temporal models
- Introduce hyperparameter tuning
- Deploy via FastAPI serverless inference layer

10. Conclusion

The Pearls AQI Predictor successfully demonstrates a complete server-less ML pipeline including:

- Automated data ingestion
- Feature engineering and storage
- Multi-model training and evaluation
- Experiment tracking and model registry
- Real-time dashboard deployment

While long-term forecasting remains a limitation, the **24-hour AQI prediction system is reliable and production-ready** within the constraints of a free-tier architecture.